DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026: Class test

Subject: Mathematics Basic (241)

SET–II

16 For the following distribution: **1**

Class	0-5	5-10	10-15	15-20	20-25
Frequency	2	4	7	5	2

Find the Mode

(a)15 (b) 17.1 (c) 18 (d)13

17 Which one of the following is true for relating mean, median and mode? **1**

(a) $mean + 2Mode = 3Median$

(b) $median = Mode + 2Mean$

(c) $Mean = (3Median - Mode)/2$

(d) $Mode + 2Median = 3Mean$

18 Getting three heads when a coin tossed thrice is **1**

(a) 1/8 (b)1 (c) 3/8 (d)1/4

19 **Assertion:** The HCF of two numbers is 18 and their product is 3072. Then their LCM = 169. **1**

Reason: If a, b are two positive integers, then HCF x LCM = a x b.

(a)Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(b)Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(c)Assertion (A) is true but reason (R) is false.

(d)Assertion (A) is false but reason (R) is true.

20 **Assertion(A):**The point(0,4) lies on y-axis. **1**

Reason(R): The x-coordinate of a point on y-axis is zero.

a)Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(b)Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(c)Assertion (A) is true but reason (R) is false.

(d)Assertion (A) is false but reason (R) is true.

SECTION B

Section B contains 5 Questions of 2 marks each

21 Solve the following pair of linear equations for x and y: **2**

$141x + 93y = 189;$

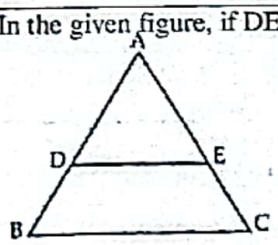
$93x + 141y = 45$

OR

Find the value of k for which the pair of linear equations $kx + y = k^2$ and $x + ky = 1$ have infinitely many solutions.

22 If $\theta = 30°$, verify that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. **2**

23 In the given figure, if DE || BC, AE = 8 cm, EC = 2 cm and BC = 6 cm, then find DE. **2**



24 Prove that tangents drawn to a circle from an external point are equal to each other. **2**

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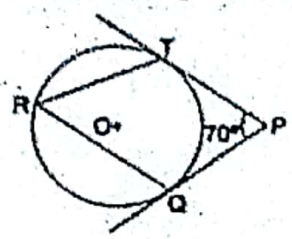
25	A chord of a circle of radius 10cm subtends a right angle at the centre. Find the Area of minor segment. (Use $\pi = 3.14$) OR A car has two wipers which do not overlap. Each wiper has a blade of length 21 cm, sweep in through an angle 120°. Find the total area cleaned at each sweep of the blades. (Use $\pi = 22/7$)	2
	SECTIONC	
	Section C contains 6 Questions of 3 marks each	
26	Prove that $\sqrt{3}$ is irrational number.	3
27	If the zeroes of the polynomial $x^2 + px + q$ are double in value to the zeroes of $2x^2 - 5x - 3$, find the value of p and q.	3
28	Find the value of k, so that the pair of linear equation has unique solution: $2x + ky = 11$ and $x - 2y = -12$	3
29	$Prove\ that\ (cosec\theta - cot\theta)^2 = \frac{1-cos\theta}{1+cos\theta}$ OR $\frac{cosA}{1+sinA} + \frac{1+sinA}{cosA} = 2secA$	3
30	In the given figure, O is the centre of a circle. PT and PQ are tangents to the circle from an external point P. If $\angle TPQ = 70°$, find $\angle TRQ$.  OR Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.	3
31	All red face cards are removed from a pack of playing cards. The remaining cards are well-shuffled and then a card is drawn at random from them. Find the probability that the drawn card is (i) a red card, (ii) a face card, (iii) a card of clubs.	3
	SECTION D	
	Section D contains 4 Questions of 5 marks each	
32	A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train. **OR** A motor boat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.	5
33	Prove that: "If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio."	5

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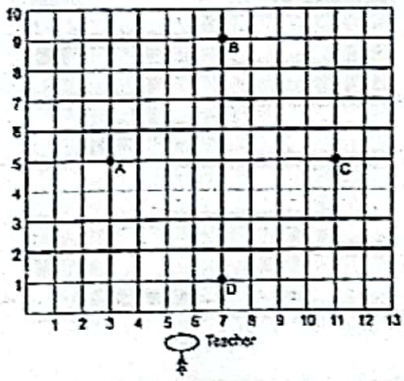
34	If the median of the distribution given below is 28.5, find the values of x and y:	5

Class interval	Frequency
0-10	5
10-20	x
20-30	20
30-40	15
40-50	y
50-60	5
Total	60

35	A juice seller was serving his customers using glasses. The inner diameter of the cylindrical glass was 5 cm, but the bottom of the glass had a hemispherical raised portion which reduced the capacity of the glass. If the height of a glass was 10 cm, find the apparent capacity of the glass and its actual capacity. (Use $\pi = 3.14$.) OR A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.	5
	SECTION E	
	Section E contains 3 Questions of 4 marks each	
36	CASE STUDY – 1 In a school, students thought of planting trees in and around the school to reduce air pollution. It was decided that each section of each class would plant twice as many plants as the class standard. There were 3 sections of each standard from 1 to 12. So, if there are three sections in class 1 say 1A, 1B and 1C, then each section would plant 2 trees. Similarly, each section of class 2 would plant 4 trees and so on. Thus, the number of trees planted by classes 1 to 12 formed an AP given by 6, 12, 18,... Based on above information answer the following question:	4
(a)	What is the common difference of the AP formed?	1
(b)	What will be the nth term of the AP formed?	1
(c)	What will be the third term from the end of the AP formed? OR How many trees will be planted by the students of all the sections of class 8?	2
37	CASE STUDY-2 Students of a school are standing in rows and columns in their playground for a drill practice. A, B, C and D are the positions of four students as shown in the figure.	4



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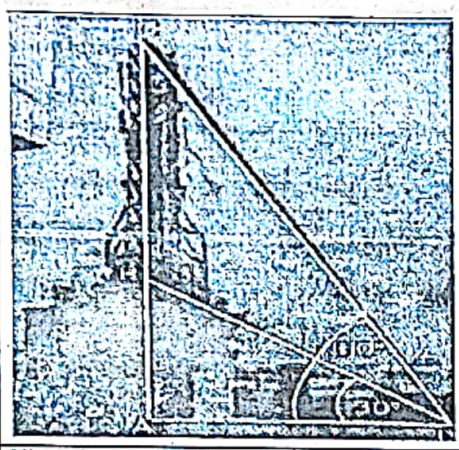
(a)	Is it possible to place Ram(R) in the drill in such a way that he is equidistant from all the four students A, B, C and D?	1
(b)	What are the coordinates of the position of Ram?	1
(c)	What are the coordinates of C and D respectively? OR What are the coordinates of A and B respectively?	2
38	**CASE STUDY 3** Consider a telecom tower BC fixed on top of a building AB. The distance between the Base of the building and point P on the ground is 48 m. From the point P the angle of elevation of the top of a building B is 30^0 and the angle of elevation of the top of the Tower C is 60^0 . 	4
(a)	Find the angle of depression from the top of the tower to the point P	1
(b)	Find the angle of depression from the top of the building to the point P.	1
(c)	What is the height of the building AB? OR Find the height of the telecom tower BC?	2

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